

a) Discuss the difference between a server and a host

Server: A server is a computer or software that provides services or resources to other devices (clients) over a network. Examples include web servers, file servers, and database servers.

Host: A host refers to any device (including servers) connected to a network that can send or receive data. It has an IP address and may host services or be a client.

Difference: All servers are hosts, but not all hosts are servers. A server provides services, while a host may simply access services or act as a general network device.

b) Define the following terms as used in Web programming

i. Data

Data refers to raw facts, figures, or input collected for processing or analysis, e.g., a user's name or age entered on a form.

ii. Database

A database is an organized collection of related data stored electronically and accessed using database management systems (DBMS), like MySQL or MongoDB.

iii. Open Database Connectivity (ODBC)

ODBC is a standard API (Application Programming Interface) that allows applications to access data from different database systems using a common set of commands, regardless of the DBMS.

iv. Variable

A variable is a named storage location in programming that holds data that can change during the execution of a program. In JavaScript, it's declared using `let`, `const`, or `var`.

c) Give three differences between client-server computing and peer to peer (p2p) computing.

Client server computing	Peer to peer computing
Relies on a central server to provide services	No central server; all peers are equal
Clients request services; servers respond	Each peer can act as both client and server
Easier to manage and secure	More difficult to manage due to decentralization
Example: Web browsing, email	Example: File sharing (e.g., BitTorrent)

d) With an example show how one can include JavaScript in HTML

Answer: JavaScript can be included in HTML using the `<script>` tag.

```
html
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<!DOCTYPE html>
<html>
<head>
  <title>JavaScript Example</title>
</head>
```

```
<body>
  <h1>Hello!</h1>
  <script>
    alert("Welcome to my website!");
  </script>
</body>
</html>
```

g) Illustrate what a function is in JavaScript (2 marks)

A function is a block of code designed to perform a specific task. It can be called whenever needed.

Example:

javascript

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```
function greetUser() {
  console.log("Hello, User!");
}
```

```
greetUser(); // Calls the function
```